THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — MATHS UNIT TEST SET B

Chapters 5–8: **Parallel and Intersecting Lines, Number Play, A Tale of Three Intersecting Lines, Working with Fractions** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–12 are Objective Type (1 mark each).
3. Q. 13–17 are Very Short Answer Type (2 marks each).
4. Q. 18–23 are Short Answer Type (3 marks each).
5. Q. 24–25 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 22 to 25.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 12 × 1 = 12

Choose the correct option and write it in your answer book.

1. Two lines in a plane that never meet are called: 1

(a) parallel lines **(b)** transversals
(c) intersecting lines **(d)** perpendicular lines

2. An angle of 90° is called a: 1

(a) right angle **(b)** reflex angle **(c)** acute angle **(d)** straight angle

3. The complement of 73° is: 1

(a) 163° **(b)** 107° **(c)** 73° **(d)** 17°

4. The Virahāṅka numbers were first studied in India while counting: 1

(a) rhythm patterns (long and short beats) **(b)** planet orbits
(c) triangles **(d)** coins

5. The sum of 7 and 8 is: 1

(a) odd **(b)** even **(c)** neither **(d)** both

6. In the sequence 2, 4, 6, 8, … the 10th term is: 1

(a) 20 **(b)** 22 **(c)** 18 **(d)** 10

7. A triangle can have at most ___ obtuse angle(s). 1

(a) one **(b)** two **(c)** three **(d)** none

8. Sides 4 cm, 4 cm and 9 cm form a triangle? 1

(a) Cannot say **(b)** Only if equal **(c)** No **(d)** Yes

9. Two angles of a triangle are 64° and 38°. The third angle is: 1

(a) 78° **(b)** 116° **(c)** 26° **(d)** 102°

10. 5/6 × 3/10 = 1

(a) 25/9 **(b)** 17/15 **(c)** 5/12 **(d)** 1/4

11. Multiplying a number by a proper fraction gives a result that is: 1

(a) always 0 **(b)** equal to the number
(c) greater than the number **(d)** smaller than the number

12. The product of a fraction and its reciprocal is: 1

(a) 2 **(b)** the fraction **(c)** 0 **(d)** 1

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

13. Define parallel and intersecting lines. 2

14. State the rule for the sum of two odd numbers and give an example. 2

15. State the angle sum property of a triangle. 2

16. Define an altitude and a median. 2

17. Divide 5/6 by 10/9. 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

18. If a transversal cuts two lines and alternate angles are 72° and 72°, what do you conclude? 3

19. Take the number 3,524. Apply the Kaprekar step once. 3

20. A triangle has angles (x + 10)°, (2x)° and (3x − 10)°. Find x and the angles. 3

21. Which is greater: 5/6 × 3/4 or 5/6 ÷ 3/4? 3

22. Find x if corresponding angles are (3x + 10)° and (5x − 30)° and lines are parallel. 3

OR

A transversal cuts two parallel lines. A corresponding angle is 65°. Find all eight angles' values. 3

23. Find the missing digit: 5,_,3 is divisible by 9 for what digit? 3

OR

Make 3 palindromes from the sum of a number and its reverse: 12 + 21 = 33; do 25 + 52. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

24. **Case study:** Two roads AB and CD are parallel and a road EF crosses them. At A the angle ∠EAB = 68°. 5

(i) Find the angle at C corresponding to it.
(ii) Find the alternate angle.
(iii) Find the co-interior angle.
(iv) Are AB and CD parallel? Why?

OR

(i) Define a transversal. 5
(ii) Name the angle pairs formed.
(iii) If a corresponding angle is 75°, find the other seven angles.

25. (i) Show the exterior angle property using a diagram. 5
(ii) If the interior opposite angles are 38° and 52°, find the exterior.
(iii) Find the third angle.

OR

(i) Construct a triangle with sides 4 cm, 5 cm, 6 cm — list the steps. 5
(ii) Draw an altitude from the 6 cm side.
(iii) Mark the orthocentre.

END OF QUESTION PAPER — Total: 50 Marks